SWAYAM NPTEL COURSE ON

STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

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Module 01: Introduction

Lecture 01: Big Data: Definition, Examples and Challenges

A. Big Data

- Definition.
- Examples.
- Challenges.

What is Big Data ?

- Data sets are too large or complex to be dealt by traditional data processing algorithms and softwares.
- Exponential growth in computing power enabled it.
- Constantly keeps getting generated.

Characteristics:

- *Volume*:

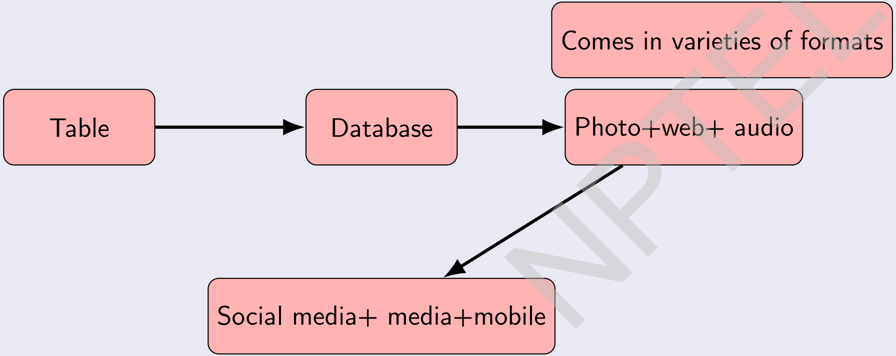
huge in size

(larger than terabytes and petabytes where
1 terabyte = 1024 gigabytes, 1 petabyte = 1,048,576 gigabytes.)

1. Immense Scale.
2. Exponential Growth.
3. Variety of Sources.
4. Need for Specialized Tools.

Characteristics (continued):

- *Variety*:



Characteristics (continued):

Types of data variety:

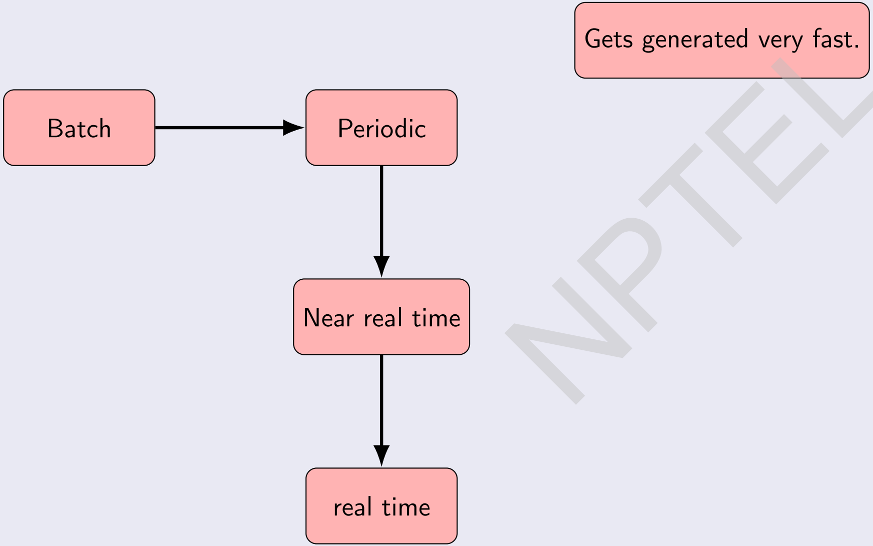
1. Structured data: Tables.
2. Semi-structured data: XML or JSON files.
3. Unstructured data: Social media posts, images, audio, and video.

Sources of variety:

1. Databases.
2. Social media.
3. Sensors.
4. Media.
5. Web logs.

Characteristics (continued):

- *Velocity*:



Characteristics (continued):

Key aspects of big data velocity:

1. Real-time processing.
2. Constant flow.
3. Importance for action.

Examples:

1. E-commerce.
2. Social media.
3. Stock trading.

Characteristics (continued):

- *Veracity*:

  Meaning of veracity :

  1. Accuracy and reliability.
  2. Data quality.
  3. Trustworthiness.

  Why this is important :

  1. Better decision-making.
  2. Risk mitigation.
  3. Improved operations.
  4. Relevant insights.

Big Data Examples

Streaming Platforms: Example - Netflix, YouTube etc.

- Analyzes: Viewars habits to provide personal recommendations.
- Data type: Video and texts.

Social Media: Example - Facebook, Twitter etc.

- Analyzes to:
  1. Find trends.
  2. User-specific recommendations.
- Data type: Network, text, audio, and video.

Supply chain optimization: Big data helps streamline supply chain operations, improve airline safety, and reduce fuel consumption.

Threat detection: Big data is used to monitor systems for vulnerabilities and detect and prevent cyber threats.

Big Data Examples (continued)

Agriculture:

- Sensors, drones, and satellite images of crop health, soil condition, and moisture to optimize irrigation and fertilization and increase crop yields.
- Data type: Images.

Weather forecasting:

- Big data for weather forecasting is created by data integration: combining satellite image + radar radiowaves + ocean buoys.
- AI and ML techniques are used to find complex models.
- Data keeps getting generated continuously and models keep improving.

Transportation:

- GPS collects satellite images.
- Using the current image and past data, predict the optimal route.

Cybersecurity:

- Computer network related data is utilized to predict cybercrime.

More Examples

- *Retail and E-commerce*: Amazon's recommendation engine analyzes browsing and purchase history to recommend products.
- *Genomic Research*: Genome sequencing projects generate vast amounts of data to advance research in personalized medicine.
- *Personalized Learning*: Online educational platforms like Coursera use big data to tailor content for individual learners.
- *Electronic Health Record*: Digitalized detailed helth records of patients create huge data sets that are used for medical research and public policy.
- *Census Data*: Data collected periodically for all citizens used for public policy.

More Examples (continued)

- *Fraud detection:* Banks monitor credit cardholders' purchasing patterns and other activity to flag atypical movements and anomalies that may signal fraudulent transactions.
- *Meteorology:* Big Data is used in
  1. Prepare weather forecasts.
  2. Predict the availability of drinking water in various world regions.
  3. Provide early warning of impending crises such as hurricanes and tsunamis.

Volume Related Problems With Big Data

1. Storage and management.
2. Performance Bottlenecks i.e, struggle to keep processing constantly flowing data quickly.
3. Processing data that is sprawled across different formats and systems.

Potential Solutions

Cloud systems, data compression, etc.

Data Quality Related Issues With Big Data

- Bad quality data is a huge challenge.
- Garbage in and Garbage out.
- Quantity and generating speed lead to flawed and/or irrelevant data.
- For example, space agencies constantly keep getting data from satellites. A huge portion of them are useless.
- Poor quality data and/or redundant data lead to a cumbersome data cleansing and validation process and wrong analysis.

Potential Solution

- Automating duplicate detection.
- Real time data monitoring tools.

Scalability Issues With Big Data

- The creation of a newer system or even a newer model cannot keep up speed-wise with the speed of data generation, so frequently existing systems/models fail to scale up.
- This slows and degrades decision-making.

Potential Solutions

1. Distributed computing.
2. Virtualized data platforms.

Other Challenges

1. Data breaches.
2. Mismanagement.
3. Security over disparate systems.
4. Complexity of data integration.
5. Cost management issues including over provisioning clouds.

References

1. Data analysis, Wikipedia.
2. Handwritten Digit Recognition with Neural Networks: From Theory to Implementation
3. kaggle dataset for height
4. Genome data set.
5. An Introduction to Statistical Methods & Data Analysis ,Seventh Edition, R. Lyman Ott, Michael Longnecker, CENGAGE Learning.

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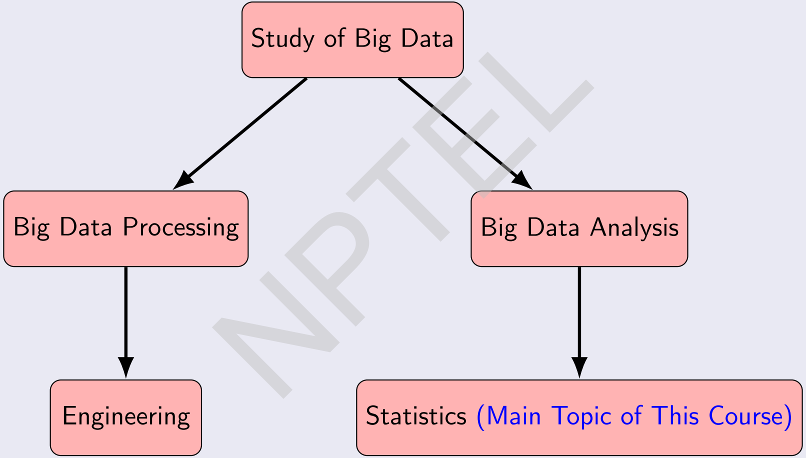
Module 01: Introduction

Lecture 02: Big Data Analysis vs Big Data Processing: Role of High-Dimensional Statistics.

Big Data Analysis vs Big Data Processing; Higher Dimensional Statistics in Big Data Analysis



Big Data Processing

- *Storing of Big Data*: Distributed file systems like
  - Google File System.
  - Hadoop.
- *Cluster computing*:
  - Nodes .
  - Networking.
  - Shared resources.
  - Parallel processing.
  - Centralized management.

Big Data Procesiing (continued)

- *Functional Programming*:
  - Scala.
  - Python.
- *Fast Data Processing on Clusters*: Spark.
- *Processing Graph Data, Stream Data etc.*

Big Data Analysis

- *Statistics*: Science behind data analysis.
- For Big Data Analysis, Statistics need to take into account the limitations of traditional systems and algorithms.
- At the end of the day, data in the Data Analysis presented in some mathematical form. Data may have many features, and Big Data Processing tends to have a massive amount of features.

Mathematical Representations of Data Points

1. ***Vectors***:
   - A tuple $(x_1, \ldots, x_n)$ of real or complex numbers.
   - Each position represents the value of some feature.
   - Examples:
     In the problem of modelling the relation between height and weight. A typical data point is (69.20477, 112.1413) a point in two dimensions.

We illustrate two examples of dataset in the next two slides.

Dataset of height vs weight

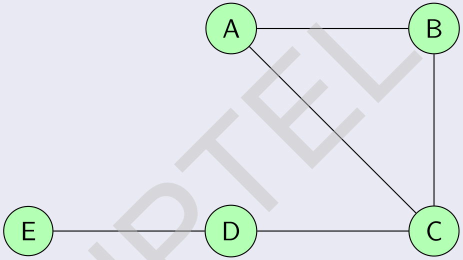
Index	Height	Weight
1	65.78331	112.9925
2	71.51521	136.4873
3	69.39874	153.0269
4	68.2166	142.3354
5	67.78781	144.2971
6	68.69784	123.3024
7	69.80204	141.4947

Dataset of genome sequence

# PC1	# PC2	# PC3	# PC4	# PC5	# PC6	# PC7
-10.90117144	0.798743345	-1.143300964	-1.070960009	11.85639581	-2.265965399	4.536404701
-9.990054297	1.416821349	-0.729626106	-0.443620754	10.41859419	0.44351371	2.640658887
-9.345388445	2.913053602	-0.921420784	0.029172626	10.67261467	-2.052552285	5.140475582
-11.2215074	1.733021061	-2.339816764	0.045786048	13.1950875	-3.068897297	2.863433874
-10.17515829	2.066307063	-0.785492616	-0.632400445	7.461272082	-1.643509092	0.715258231
-13.31992242	2.737273362	-1.532659031	0.058068544	-2.660751875	-0.550663024	2.141832367
-10.44169213	0.872937537	-0.100213953	0.519791326	0.094731956	0.764297776	-4.658872392
-9.218514509	1.758892098	-0.578552353	-0.937131202	1.373940452	-1.62012629	0.919812475
-12.673631	3.015143961	-2.100390149	-0.517233089	1.116705887	1.096178869	-3.240629907
-8.342017433	3.451908119	-1.6689321559999999	-0.158435218	-1.710343582	0.813721929	-4.333145837
-10.00354586	5.123690809	-0.941167859	-0.107404688	0.954424411	-3.093075065	-0.783159977
-9.927064519	1.051759512	-3.225099329	-1.296031613	-3.038653364	4.477170342	0.789175627
-10.25617439	3.971784407	-2.110359934	-0.180311866	-0.622544214	1.981484299	1.543862414

Mathematical Representations of Data Points (continued)

2. ***Graphs***: Vertex + edges



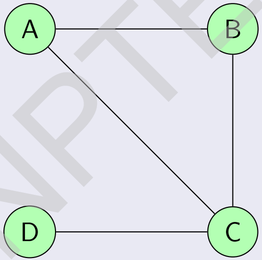
- Directed/ undirected.
- Unweighted/ Vertex-Weighted/ Edge-Weighted / Vertex + Edge-Weighted.
- Examples:

Undirected graph of k- distant friendship neighbourhood graph of Facebook profiles, i.e., each point is the graph of profiles that are k-distance from a fixed profile.

Graphical representation of the previous example

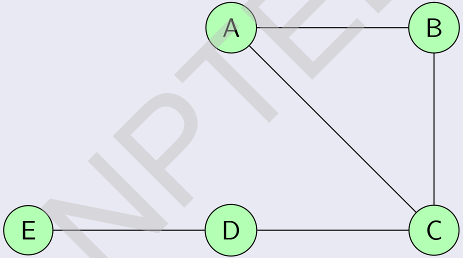
Let $k = 2$.

- For the vertices A and B the corresponding graph would be



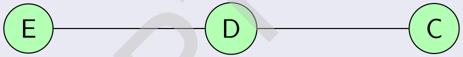
Graphical representation of the previous example (continued)

- For the vertices C, D the corresponding graph would be the whole graph i.e.,



Graphical representation of the previous example (continued)

- For the vertex E, the corresponding graph would be the whole graph i.e.,



Example (continued)

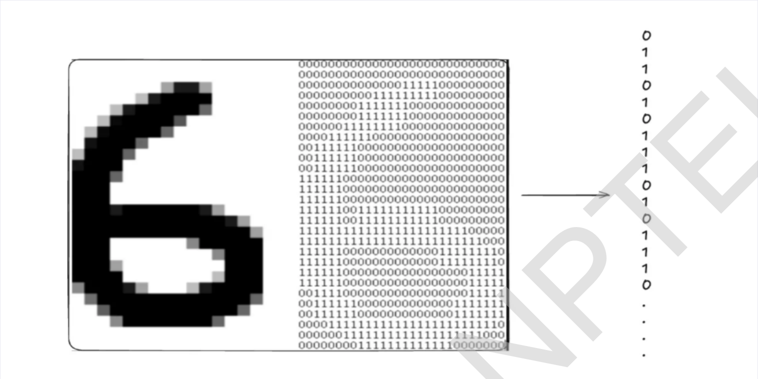
- *Weighted version*: Same graph as above with each node having weight as its number of immediate friends.

Mathematical Representations of Data Points (continued)

3. ***Matrix***: Each data point is a matrix like $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

Mathematical Representations of Data Points (continued)

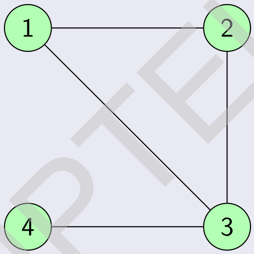
- Example: Data set consisting of the pixel images of a fixed size.



- Every matrix of order $m \times n$ can be flattened to a vector of $\mathbb{R}^{mn}$. For instant $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ can be treated as the vector $(1, 2, 3, 4, 5, 6)$ of $\mathbb{R}^6$.
- With every graph, we can associate a symmetric matrix called "adjacency matrix of the graph". If the vertex i is connected with j, then the ij-th entry of the matrix $a_{ij} = 1$, otherwise $a_{ij} = 0$.

Mathematical Representations of Data Points (continued)

For example, if the graph is



then the adjacency matrix is $\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

Mathematical Representations of Data Points (continued)

- This enables us to analyze the graph or matrix data as vector data.
- For Big Datas, the number of features is extremely high.
- Multivariate statistics of extremely high dimension is needed.

References

1. Data analysis, Wikipedia.
2. Handwritten Digit Recognition with Neural Networks: From Theory to Implementation
3. kaggle dataset for height
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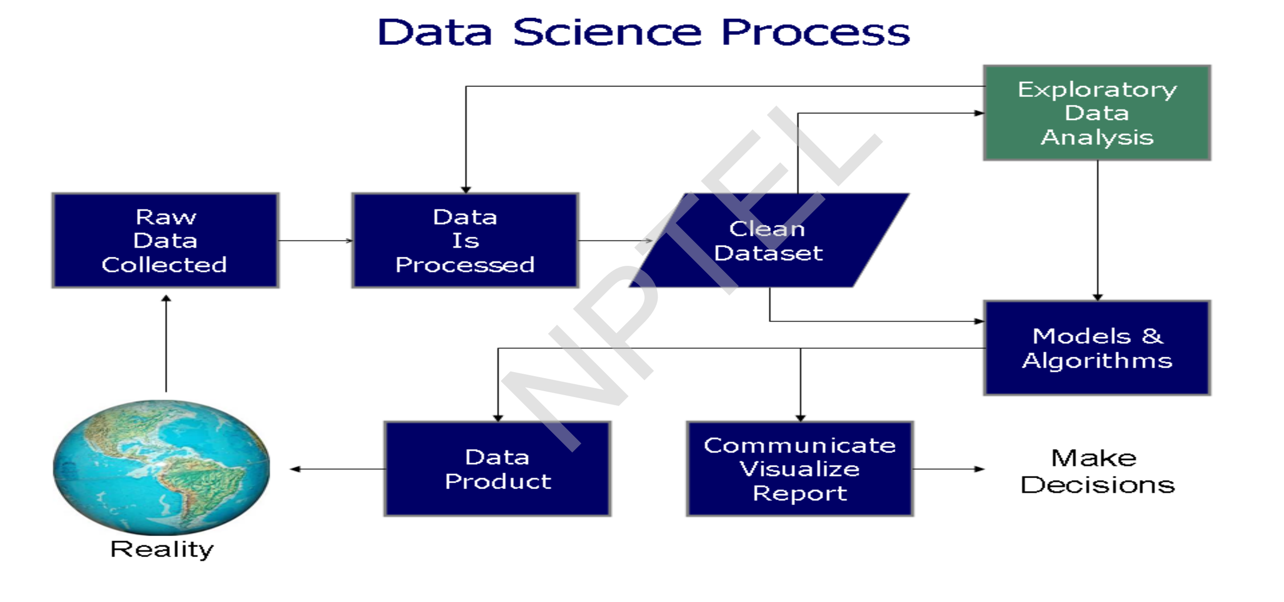
Module 01: Introduction

Lecture 03: Statistical Framework 1

Data Science Process



Definition (John Tukey)

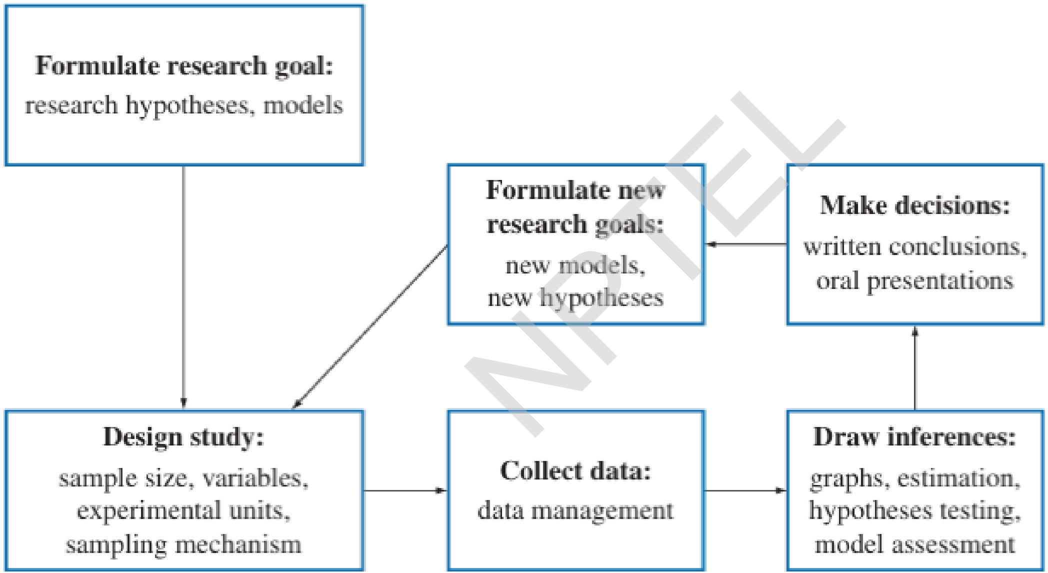
Procedures for analyzing data, techniques for interpreting the results of such procedures, ways of planning the gathering of data to make its analysis easier, more precise or more accurate, and all the machinery and results of (mathematical) statistics which apply to analyzing data.

In simple terms, statistics is the science of learning from data.

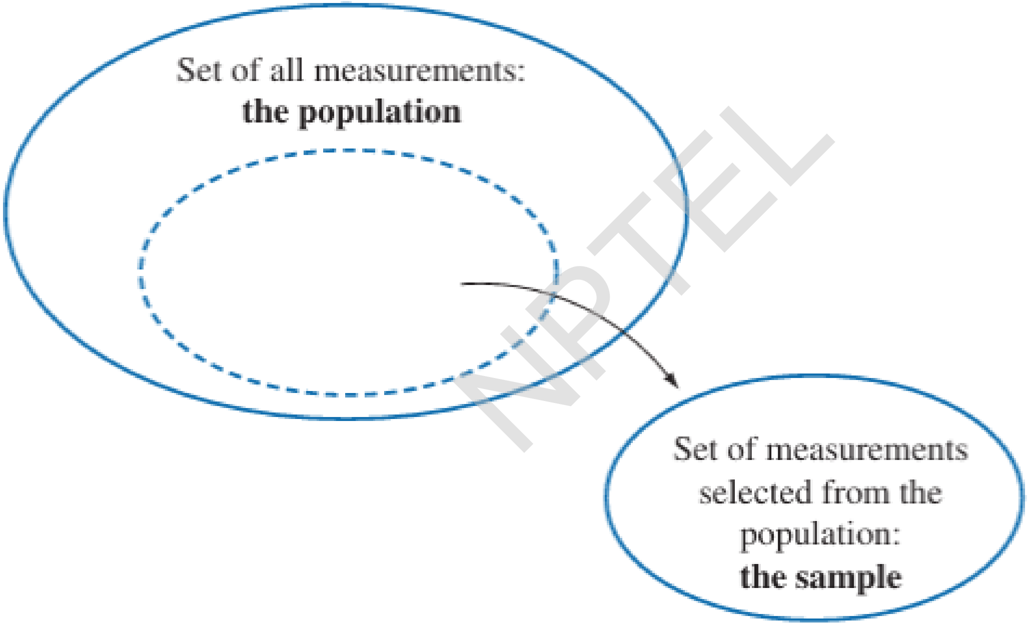
Definitions

- A **population** is the set of all measurements of interest to the sample collector.
- A **sample** is any subset of measurements selected from the population.

Scheme of Scientific Research



Statistical Method

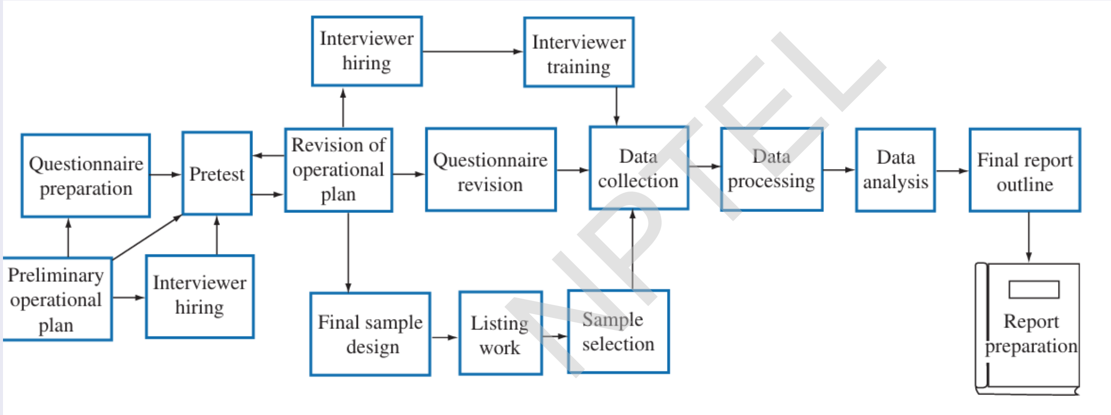


Step 1: Data Collection

The following methods are used to collect data.

- *Surveys and questionnaires:* Collecting data from a sample of people using a set of questions.
- *Interviews:* Conducting one-on-one conversations with individuals to gather in-depth information.
- *Observations:* Systematically watching and recording the behavior of subjects in their natural environment.
- *Focus groups:* Gathering a small group of people to discuss a specific topic, led by a moderator.
- *Experiments:* Manipulating variables to test a hypothesis.
- *Documents and records:* Using previously compiled information from sources like books, journals, or company reports.
- *Archival research:* Analyzing data that has already been collected and archived by other organizations.

Chart For Data Collection



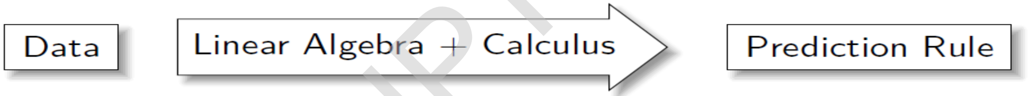
Step 2: Data Summary and Visualization

- **Data Summary:**
  - The *mode* of a set of measurements is defined to be the measurement that occurs most often .
  - The *median* of a set of measurements is defined to be the middle value when the measurements are arranged from lowest to highest.
  - The *arithmetic mean*, or *mean*, of a set of measurements is defined to be the sum of the measurements divided by the total number of measurements. The mean of $y_1, \ldots, y_n$ is $\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$.
  - The *variance* of n measurements $y_1, \ldots, y_n$ with mean $\bar{y}$ is defined as $\frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n}$. The positive square root of variance is called *standard deviation.*
- **Data Visualization:**
  - Pie Chart.
  - Bar Chart.
  - Histogram.
  - Gantt Chart.
  - Heat Map.
  - Box and Whisker Plot.
  - Waterfall Chart.
  - Area Chart.
  - Scatter Plot.
  - Pictogram Chart.

Step 3: Data Analysis

- Inferences About Population Central Values.
- Inferences Comparing Two Population Central Values.
- Inferences About Population Variances.
- Categorical data analysis.
- Linear and Logistic Regression and Correlation.
- Multiple Regression and the General Linear Model.
- Analysis of variance and covariance.

Statistics



Machine Learning

$\boxed{\text{Big Data}}$ $\xrightarrow{\text{LA + C + HPC}}$ $\boxed{\text{Prediction Rule}}$

- LA=Linear Algebra
- C=Calculus
- HPC=High Performance Computation

Statistics

Statistics In Technical Terms

Probability Distribution → Experiment → Probability Values

Example

Coin With Head Probability p → n Tosses → $\hat{p}$ = no of heads/n

- p unknown
- $\hat{p}$ is the prediction about p
- Here population is the totality of all possible tosses
- Data is the n tosses that are performed

Statistics

Data $\rightarrow$ Linear Algebra + Calculus $\rightarrow$ Prediction Rule

References

1. Data analysis, Wikipedia.
2. Handwritten Digit Recognition with Neural Networks: From Theory to Implementation
3. kaggle dataset for height
4. Genome data set.
5. Methods-of-data-collection
6. Data collection methods
7. Data Visualization geeksforgeeks
8. Datacamp
9. Springboard
10. An Introduction to Statistical Methods & Data Analysis ,Seventh Edition, R. Lyman Ott, Michael Longnecker, CENGAGE Learning.

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Module 01: Introduction

Lecture 04: Statistical Framework 2

Random Variable: Framework

- A random variable X is a function that can take all possible values of some measurements.
- So for a random variable X and a set $A \subset \mathbb{R}$, there exists an inherent probability function $P(X \in A)$.
- A random vector X takes all possible values of some measurement yielding n- dimensional vector values and $\forall\ B \in \mathbb{R}^n$ we have $P(X \in B)$. This P is called the *probability distribution* of X of size n.
- A sample generally n random variables $X_1, \ldots, X_n$ having the same distribution P (representing some underlying population X).
- For some random variable X, the *Cumulative Distribution Function* function or cdf is $F_X(a) = P(X \leq a)$ for $a \in \mathbb{R}$.
- For some random vector vector $\underline{Y} = (Y_1, \ldots, Y_n)$ the cdf $F_{\underline{Y}}(a_1, \ldots, a_n) := P(Y_1 \leq a_1, \ldots, Y_n \leq a_n)\ \forall\ (a_1, \ldots, a_n) \in \mathbb{R}^n$.

Definition

- A random variable X is called a *discrete random variable* if X takes values on a countable set $R_X = \{x_1, x_2, x_3, \ldots\}$. The *probability mass function* of a discrete random variable is given by:

$$P(X = x) = \begin{cases} p_j, & x = x_j \in R_X \\ 0, & \text{otherwise} \end{cases}$$

- For a discrete probability distribution X, the r^{th} moment of X is given by:

$$E(X^r) = \sum_{x \in R_x} x^r P(X = x)$$

In particular, the *mean* of the random variable X is given by $E(X) = \sum_{x \in R_x} xP(X = x)$ and the *variance* of X is given by

$Var(X) = E(X^2) - (E(X))^2$.

Definition (Continued)

Finally, the *moment generating function* (MGF) of X is given by

$$E(e^{tX}) = \sum_{x \in R_x} e^{tx} P(X = x).$$

Also $E(X^k) = \frac{d^k}{dt^k}(M_X(t))|_{t=0}$.

Some Standard Discrete Probability Distributions

Discrete Uniform Distribution

- Notation: $X \sim Unif(n)$.
- Probability mass Function: $P(X = x) = \frac{1}{n}, \ x = 1, 2, \ldots, n$
- Parameters : $n \in \mathbb{N}$ (number of equally likely outcomes).
- Mean: $E(X) = \frac{n+1}{2}$.
- Variance: $Var(X) = \frac{n^2-1}{12}$.
- MGF: $\frac{1-e^{tn}}{n(1-e^t)}$.

Calculation of $E(X)$ and $Var(X)$

- $E(X) = \sum_{i=1}^{n} i \frac{1}{n}$
  $= \frac{(n+1)}{2}$.
- $E(X^2) = \sum_{i=1}^{n} i^2 \frac{1}{n}$
  $= \frac{(n+1)(2n+1)}{6}$.
  Therefore $Var(X) = \frac{n^2-1}{12}$.

Poisson Distribution

- Notation: $X \sim Pois(\lambda)$
- Probability mass Function: $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \; x = 0, 1, 2, \ldots$
- Parameters : $\lambda \in (0, \infty)$ (rate).
- Mean: $E(X) = \lambda$.
- Variance: $Var(X) = \lambda$.
- MGF: $e^{\lambda(e^t - 1)}$.

Calculation of $E(X)$ and $Var(X)$

- $E(X) = \sum_{x\geq 0} x\frac{1}{x!}\lambda^x e^{-\lambda}$
  $= \lambda e^{-\lambda}\sum_{x\geq 1}\frac{1}{(x-1)!}\lambda^{x-1}$
  $= \lambda e^{-\lambda}\sum_{j\geq 0}\frac{\lambda^j}{j!}$
  $= \lambda e^{-\lambda}e^{\lambda}$
  $= \lambda$
- $E(X^2) = \sum_{x\geq 0} x^2\frac{1}{x!}\lambda^x e^{-\lambda}$
  $= \lambda e^{-\lambda}\sum_{x\geq 1}\frac{x}{(x-1)!}\lambda^{x-1}$
  $= \lambda e^{-\lambda}(\sum_{x\geq 1}(x-1)\frac{1}{(x-1)!}\lambda^{x-1} + \sum_{x\geq 1}\frac{1}{(x-1)!}\lambda^{x-1})$
  $= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda})$
  $= \lambda(\lambda + 1)$.

  $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$

Binomial Distribution

- Notation $X \sim B(n, p)$.
- Probability mass Function: $P(X = x) = {}^nC_x p^x (1-p)^{n-x}, \; x = 0, 1, 2, \ldots, n$
- Parameters :
  1. $n \in \{0, 1, 2, 3, \ldots\}$ (number of trials).
  2. $p \in [0, 1]$ (success probability of each trial).
- x in the number of success.
- Mean: $E(X) = np$.
- Variance: $Var(X) = np(1-p)$.
- MGF: $((1-p)pe^t)^n$.

Geometric Distribution

- Notation: $X \sim Geom(p)$.
- Probability Mass Function: $P(X = x) = p(1-p)^x, \; x = 0, 1, 2, \ldots$
- Parameter : $p \in [0, 1]$ (success probability of each trial).
- x is the number of failures before one success.
- Mean: $E(X) = \frac{(1-p)}{p}$.
- Variance: $Var(X) = \frac{(1-p)}{p^2}$.
- MGF: $\frac{p}{1-(1-p)e^t}$, for $t < -log(1-p)$.

Negative Binomial Distribution

- Notation $X \sim NB(r, p)$.
- Probability mass Function: $P(X = x) = {}^{x+r-1}C_x p^r (1-p)^x, \;\; x = 0, 1, 2, \ldots, n$
- Parameters :
  1. $r > 0$ (number of successes until experiment is stopped).
  2. $p \in [0, 1]$ (success probability of each trial).
- x deontes the number of failure.
- Mean: $E(X) = \frac{r(1-p)}{p}$.
- Variance: $Var(X) = \frac{r(1-p)}{p^2}$.
- MGF: $(\frac{p}{1-(1-p)e^t})^r$, for $t < -log(1-p)$

Hyper-geometric Distribution

- Notation: $X \sim HG(N, K, n)$.
- Probability mass Function: $P(X = x) = \frac{{}^{K}C_x^{N-K}C_{n-x}}{{}^{N}C_n}$, $x \in \{\max(0, n + K - N), \ldots, \min(n, K)\}$
- Parameters :
  1. $N \in \{0, 1, 2, \ldots\}$ (finite population size).
  2. $K \in \{0, 1, 2, \ldots, N\}$ (objects with the desired feature).
  3. $n \in \{0, 1, 2, \ldots, N\}$ (number of draws).
- x is the number of observed success .
- Mean: $E(X) = \frac{nK}{N}$.
- Variance: $Var(X) = \frac{nK(N-k)(N-n)}{N^2(N-1)}$.
- MGF: Not an elementary function.

Definiton

A random variable X is called an *(absolutely) continuous random variable* if there exists a non-negative funtion $f(x)$, defined for all real $x \in (-\infty, \infty)$, having the property for any set S of real numbers,

$$\mathbb{P}(X \in S) = \int_S f(x)dx.$$

The function $f(x)$ is called the *probability density function* or *pdf* of X.

1. $1 = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x)dx.$
2. If $S = [a, b]$, then $P(X \in S) = \int_a^b f(x)dx$. In particular if $a = b$, then

$$P(X = a) = \int_a^a f(x)dx = 0.$$

Definiton (continued)

Let X be a continuous random variable having pdf $f(x)$. We have the following definitions below:

1. The *expected value* or the *mean* of X, is defined as

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

2. If r is a natural number, then the *r-th moment* of X is defined as

$$E(X^r) = \int_{-\infty}^{\infty} x^r f(x)dx.$$

3. The *cumulative distribution function* or *cdf* of X is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)dx.$$

4. The *moment generating function* (MGF) of X is given by

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)dx.$$

Also in case of continuous distribution, $M_X(t) = E(X^k) = \frac{d^k}{dt^k}(M_X(t))|_{t=0}$.

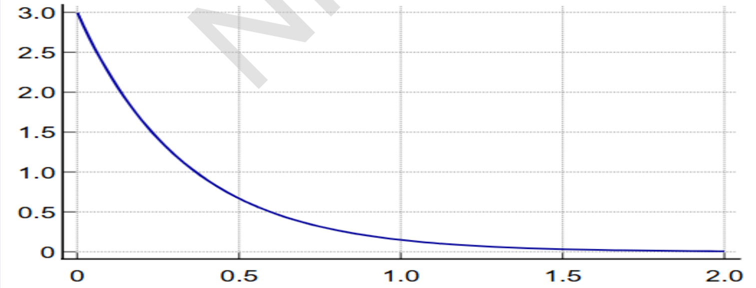
Some Continuous Probability Distributions

Exponential distribution

- PDF:

$$f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta} & \text{if } 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

  where $\beta > 0$.
- Parameters: β (inverse rate).
- Mean : $E(X) = \beta$.
- Variance : $Var(X) = \beta^2$.
- MGF: $(1 - \beta t)^{-1}$ for $t < \beta^{-1}$.



Calculation of $E(X)$ and $Var(X)$

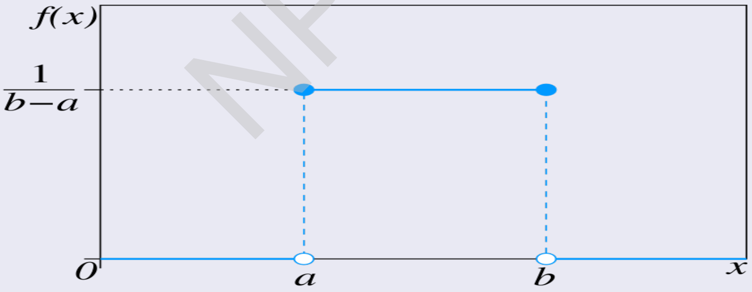
- $E(X) = \frac{1}{\beta}\int_0^\infty xe^{-x/\beta}dx$
  $= \frac{1}{\beta}\lim_{B\to\infty}\int_0^B xe^{-x/\beta}dx$
  $= \frac{1}{\beta}\lim_{B\to\infty}(-\beta Be^{-B/\beta} + \beta\int_0^B e^{-x/\beta})dx$
  $= \beta$
- In a similar manner, $E(X^2) = 2\beta^2$.
  Thus, $Var(X) = E(X^2) - (E(X))^2$
  $= \beta^2$.

Uniform distribution on the interval $[a, b]$ where $-\infty < a < b < \infty$

- Notation: $X \sim \mathcal{U}_{[a,b]}$
- PDF:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0, & \text{otherwise.} \end{cases}$$

- Mean: $E(X) = \frac{a+b}{2}$.
- Variance: $Var(X) = \frac{(b-a)^2}{12}$.
- MGF $= \frac{1}{t(b-a)}(e^{tb} - e^{ta}), \qquad t \neq 0.$

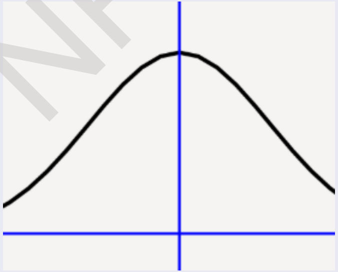


Calculation of $E(X)$ and $Var(X)$

- $E(X) = \int_a^b \frac{x}{b-a} dx$
$= \frac{1}{b-a}[\frac{x^2}{2}]_a^b$
$= \frac{(b^2-a^2)}{2(b-a)}$
$= \frac{a+b}{2}$.
- $E(X^2) = \int_a^b \frac{x^2}{b-a} dx$
$= \frac{1}{b-a}[\frac{x^3}{3}]_a^b$
$= \frac{(a^2+ab+b^2)}{3}$.
Therefore, $Var(X) = E(X^2) - (E(X))^2$
$= \frac{(a^2+ab+b^2)}{3} - \{(\frac{a+b}{2})\}^2$
$= \frac{(b-a)^2}{12}$

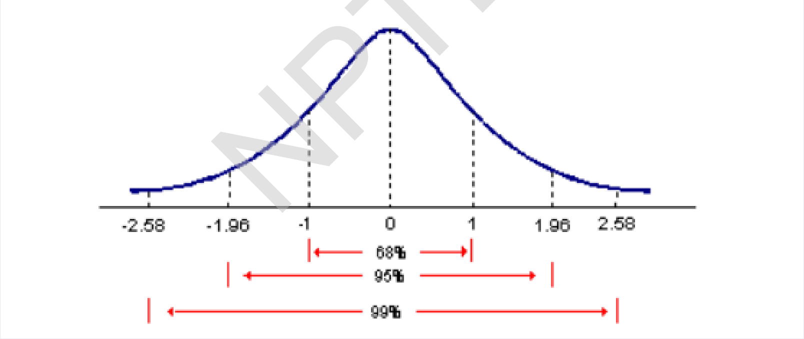
Normal distribution

- Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$.
- PDF: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $x \in (-\infty, \infty)$
- Parameters:
  1. μ (location)
  2. $\sigma^2 > 0$ (scale).
- Mean: $E(X) = \mu$,
- Variance: $Var(X) = \sigma^2$.
- MGF: $e^{(\mu t + \frac{\sigma^2 t^2}{2})}$



Standard Normal Distribution

- Notation: $X \sim \mathcal{N}(0,1)$
- PDF: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \; x \in (-\infty, \infty)$
- Mean: $E(X) = 0.$
- Variance : $Var(X) = 1.$
- MGF: $e^{\frac{t^2}{2}}.$

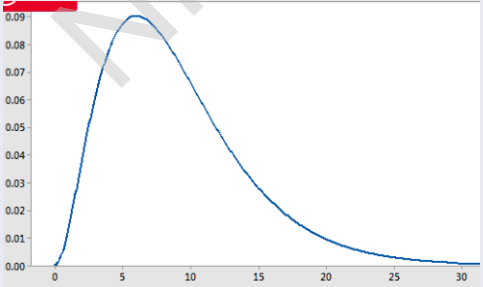


Gamma distribution

- PDF:

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & \text{if } 0 < x < \infty \\ 0, & \text{if } x \leq 0 \end{cases}$$

  where α, β are positive real numbers and $\Gamma(\alpha) = \int_0^\infty e^{-x} x^{\alpha-1} dx$.
- Parameters:
  1. $\alpha > 0$ (shape).
  2. $\beta > 0$ (scale).
- Mean: $E(X) = \alpha\beta$
- Variance $Var(X) = \alpha\beta^2$.
- MGF: $(1-\beta t)^{-\alpha}, \quad t < \frac{1}{\beta}$.

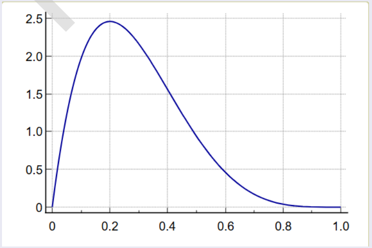


Beta distribution

- Notation: $X \sim \text{Beta}\ (\alpha, \beta)$.
- PDF: $f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$ where, $\alpha, \beta > 0$ and $0 < x < 1$ and

$$B(\alpha, \beta) = \int_{0+}^{1-} x^{\alpha}(1-x)^{\beta-1} dx.$$

- Parameters :
  1. $\alpha > 0$ (shape),
  2. $\beta > 0$ (shape).
- Mean: $E(X) = \frac{\alpha}{\alpha+\beta}$
- Variance: $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.
- MGF $= \sum_{j=0}^{\infty} \frac{t^j}{\Gamma(j+1)} \frac{\Gamma(\alpha+j)\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+j)\Gamma(\alpha)}$.

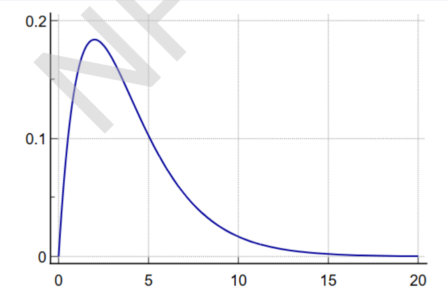


Chi-square distribution with n-degrees of freedom

- Notation: $X \sim \chi^2_n$.
- PDF:

$$f(x) = \begin{cases} \frac{1}{\Gamma(n/2)2^{n/2}} e^{-x/2} x^{n/2-1} & \text{if } 0 < x < \infty \\ 0, & \text{if } x \leq 0 \end{cases}$$

- Parameter : n.
- Mean : $E(X) = n$.
- Variance : $Var(X) = 2n$.
- MGF$=(1-2t)^{-n/2}$ for $t < \frac{1}{2}$.



F-distribution with (m, n)-degrees of freedom

Let X and Y be independent χ^2 random variables, with m and n degrees of freedom respectively. Then the random variable

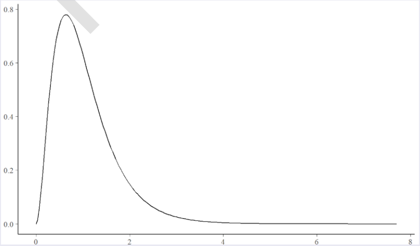
$$F = \frac{X/m}{Y/n}$$

is said to have an *F-distribution with* (m, n) degrees of freedom.

- PDF:

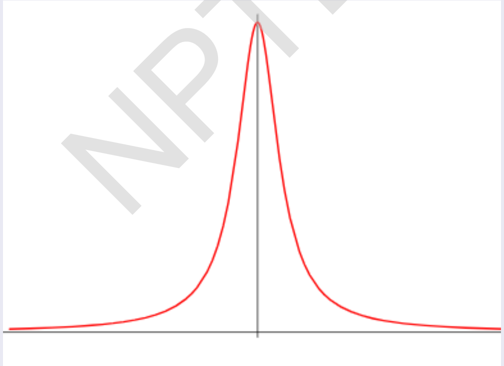
$$g(f) = \begin{cases} \frac{\Gamma[(m+n)/2]}{\Gamma(m/2)\Gamma(n/2)}\left(\frac{m}{n}\right)\left(\frac{m}{n}f\right)^{(m/2)-1}\left(1+\frac{m}{n}f\right)^{-(m+n)/2} & \text{if } f > 0 \\ 0, & \text{if } f \leq 0 \end{cases}$$

- Parameters : $m, n > 0$.
- Mean : $E(X) = \frac{n}{n-2}, \quad n > 2.$
- Variance: $Var(X) = \frac{n^2(2m+2n-4)}{m(n-2)^2(n-4)}, \quad n > 4.$



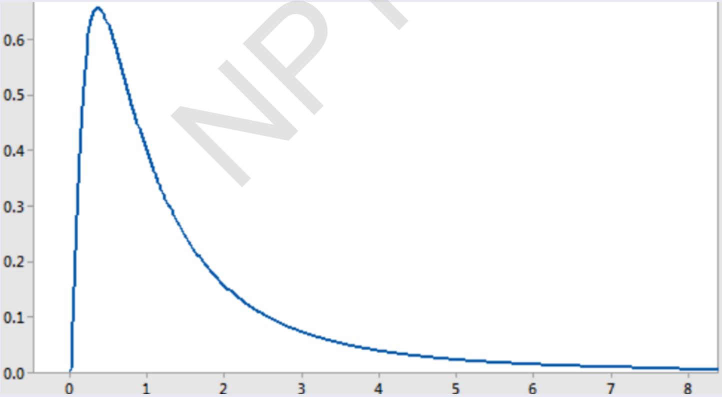
Cauchy distribution

- PDF: $f(x) = \frac{\mu}{\pi}\frac{1}{\mu^2+(x-\theta)^2}, \quad \infty < x < \infty, \mu > 0.$
- Parameters:
  1. $\mu > 0$ (scale),
  2. θ(location).
- Mean: undefined.
- MGF: Moments of order less than 1 exist, but the moments of order greater than or equal to 1 do not exist.



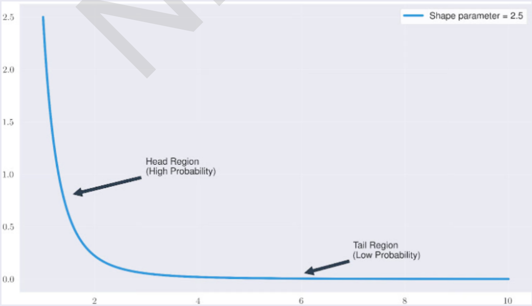
Log-Normal Distribution

- PDF: $f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln(x)-\mu)^2}{2\sigma^2}\right), \quad x > 0$
- Parameters:
  1. Logarithm of Location, $\mu \in (-\infty, \infty)$
  2. Logarithm of Scale $\sigma > 0$.
- Mean: $E(X) = e^{\left(\mu+\frac{\sigma^2}{2}\right)}$,
- Variance: $Var(X) = e^{(2\mu+\sigma^2)}(e^{\sigma^2} - 1)$,
- MGF: Not defined for any positive value of the argument t.



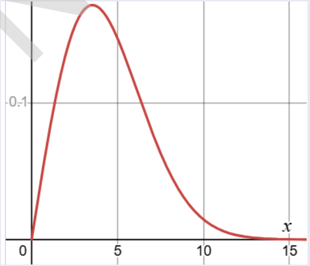
Pareto Distribution

- Notation $X \sim Pa(\alpha, x_m)$.
- PDF: $f(x) = \dfrac{\alpha x_m^{\alpha}}{x^{\alpha+1}}$
- Parameters :
  1. Scale, $x_m > 0$.
  2. Shape, $\alpha > 0$.
- Mean: $E(X) = \begin{cases} \infty, & \alpha \le 1, \\ \frac{\alpha x_m}{\alpha - 1} & \alpha > 1. \end{cases}$,
- Variance: $Var(X) = \begin{cases} \infty, & \alpha \le 2, \\ \frac{x_m^2 \alpha}{(\alpha-2)(\alpha-1)^2} & \alpha > 2. \end{cases}$
- MGF: Does not Exist



Weibull Distribution

- Notation $X \sim Weibull(\lambda, k)$.
- PDF: $f(x) = \begin{cases} \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-\left(\frac{x}{\lambda}\right)^k}, & \text{when } x \geq 0, \\ 0, & \text{when } x < 0. \end{cases}$
- Parameters :
  1. Scale, $\lambda \in (0, \infty)$.
  2. Shape, $k \in (0, \infty)$.
- Mean: $E(X) = \lambda\Gamma(1 + 1/k)$,
- Variance: $Var(X) = \lambda^2 \left(\Gamma\left(1 + \frac{2}{k}\right) - \left(\Gamma\left(1 + \frac{1}{k}\right)\right)^2\right)$
- MGF: $\sum_{n=0}^{\infty} \frac{t^n \lambda^n}{n!} \Gamma(1 + n/k)$ where $k \geq 1$.



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SWAYAM NPTEL COURSE ON

STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 01: Introduction

Lecture 05: Statistical Framework 3

Statistical Framework

1. Recall that
   - A **population** is the set of all measurements of interest to the sample collector.
   - A **sample** is any subset of measurements selected from the population.
2. *Goal :* Want to infer about population using sample.
3. As population is the set of "all possible" measurements, choice of a particular or particular set of measurement is in the domain of chances.
4. This takes us to the domain of probability.

Statistical Framework (continued)

Particularly,

- Probability of picking a set A of measurements out of the population X (“all possible measurement”) is a number $0 \leq P(A) \leq 1$.
- P(nothing)=0, P(everything)=1.
- This function P intrinsic to X and A.
- One important goal of statistics is to develop methods to understand this P using sample.

Statistical Framework (continued)

- If $A_1, \ldots, A_k$ are disjoint then $P(\cup_{i=1}^{k} A_i) = \sum_{i=1}^{k} P(A_i)$
- $P(A^c) = 1 - P(A)$.

Example 1

For a given coin, we want to understand the chance of getting head.

- If the chance is p for $p \in [0, 1]$, then tossing the coin large amount of time will yield approximately $100p\%$ heads.
- *Goal:* Guess p.
- *Population:* All possible tosses of that coin.
- *Sample Collection:* Toss the coin n times. Let

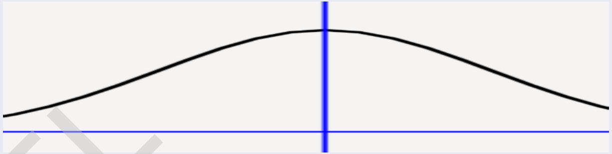
$$X_i = \begin{cases} 1 & \text{if } i\text{-th toss is a head} \\ 0 & \text{otherwise.} \end{cases}$$

Example 1 (continued)

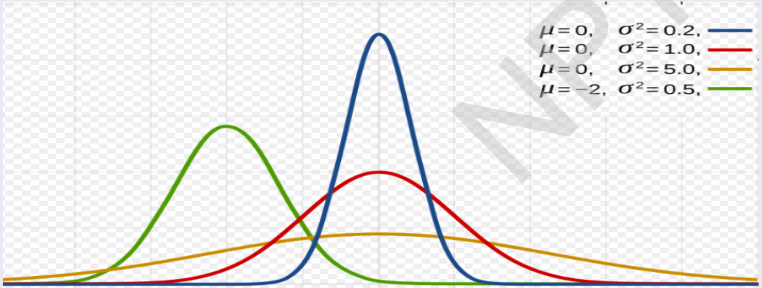
- A "decent" guess for p will be $\hat{p} = \frac{\sum X_i}{n}$.
- Sample size n is important as for large n, $\hat{p} \approx p$.
- Question:

  For a fixed n, how good an approximation $\hat{p}$ is ?

Example 2



- Heights of all Indian people is known to follow "Bell curve".
- Want to produce the average height based on heights of n people $H_1, \ldots, H_n$. So population: All possible height, sample : $H_1, \ldots, H_n$.
- Also want to predict how "spread out" the heights are.



Therefore, according to the color orange > red > green >blue.

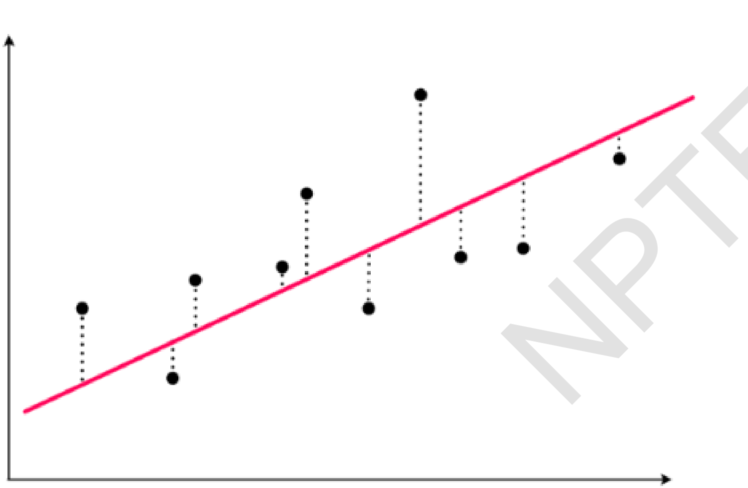
Example 2 (continued)

- Measure of center: Mean.
- Measure of spread: Variance.
  For values $X_1, \ldots, X_n$, Mean $= \frac{\sum_{i=1}^{n} X_i}{n} = \bar{X}$, Variance $= \frac{(X_i - \bar{X})^2}{n}$.
- Guess the population average: $\bar{H} = \frac{\sum_{i=1}^{n} H_i}{n}$.
- Guess the population variance: $\frac{\sum_{i=1}^{n}(H_i - \bar{H})^2}{n}$.
- Fun fact: $\frac{\sum_{i=1}^{n}(H_i - \bar{H})^2}{n-1}$ is also a good guess, as we shall see soon!

Example 3

- For some fixed population of people want to predict body weight in terms of height.
- *Population:* (height, weight) pairs of all people.
- *Sample :* Measured (height, weight) pairs of n people $(x_1, y_1), \ldots, (x_n, y_n)$.
- Want to find a nice function, $y = f(x)$.
- One decent guess of f: the least square line !!
- Least Square Line: Line $y = \hat{a} + \hat{b}x$ such that $a = \hat{a}$ and $b = \hat{b}$ minimizes $\sum_{i=1}^{n}(y_i - a_i - bx_i)^2$

Example 3 (continued)



Example 3 (continued)

Finding $\hat{a}$ and $\hat{b}$.

- Let $S = S(a,b) = \sum_{i=1}^{n}(y_i - a - bx_i)^2$.
- Minimize S.
- $\frac{\partial S}{\partial a} = -2\sum_{i=1}^{n}(y_i - a - bx_i)$ and $\frac{\partial S}{\partial b} = -2\sum_{i=1}^{n}(y_i - a - bx_i)x_i$.
- Solution is to find (a,b) such that $\frac{\partial S}{\partial a} = \frac{\partial S}{\partial b} = 0$
- The estimates are $\hat{a}$ of a and $\hat{b}$ of b.
- $\hat{a} = \bar{y} - \hat{b}\bar{x}$
- $\hat{b} = \frac{s_{xy}}{s_{xx}}$
  where $s_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}),\ s_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$.

Example 3 (continued)

- $\frac{\partial^2 S}{\partial a^2} = -2\sum_{i=1}^{n}(-1) = 2n$, $\frac{\partial^2 S}{\partial b^2} = 2\sum_{i=1}^{n} {x_i}^2$, $\frac{\partial^2 S}{\partial a \partial b} = 2\sum_{i=1}^{n} = 2n\bar{x}$
- The Hessian matrix

$$H^* = \begin{bmatrix} \frac{\partial^2 S}{\partial a^2} & \frac{\partial^2 S}{\partial a \partial b} \\ \frac{\partial^2 S}{\partial a \partial b} & \frac{\partial^2 S}{\partial b^2} \end{bmatrix}$$

i,e.,

$$H^* = \begin{bmatrix} n & n\bar{x} \\ n\bar{x} & \sum_{i=1}^{n} x_i \end{bmatrix}$$

- $\det(H^*)$
  $= 4n\sum_{i=1}^{n}(x_i - \bar{x})^2 \geq 0$. The case when $\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0$, all observations are identical and this case is not interesting. Therefore, hence $\det(H^*) > 0$ and S has a minimum at $(\hat{a}, \hat{b})$.
- Observe that as the $\hat{a}$ and $\hat{b}$ depend on the sample i.e., specific measurements out of all possible measurements, their values too are "probabilistic" i.e., they take certain values with certain probabilities.

Worked out problem

Calculate Fitting straight line - Curve fitting using Least square method

Table: Sample Data Table

X	Y
5	1
4	2
3	3
2	4
1	5

Solution

Straight line is $y = \hat{b}x + \hat{a}$ where

$$\hat{b} = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n}(x^2) - (\sum_{i=1}^{n} x)^2}$$

$$\hat{a} = \frac{\sum_{i=1}^{n} y - m\sum_{i=1}^{n} x}{n}$$

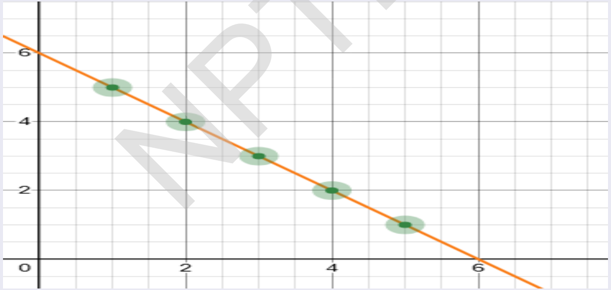
Table: Sample Data Table

x	y	x^2	xy
5	1	25	5
4	2	16	8
3	3	9	9
2	4	4	8
1	5	1	5
$\sum x = 15$	$\sum y = 15$	$\sum x^2 = 55$	$\sum xy = 35$

Solution

$\hat{b} = \frac{5 \cdot 35 - 15 \cdot 15}{5 \cdot 55 - (15)^2}$
$= -1$

$\hat{a} = \frac{15 - (-1) \cdot 15}{5}$
$= 6$
So, the required equation is $y = -x + 6$.



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Thank you